

9.3.17.

- a. How many strings of four hexadecimal digits do not have any repeated digits?
- b. How many strings of four hexadecimal digits have at least one repeated digit?
- c. What is the probability that a randomly chosen string of four hexadecimal digits has at least one repeated digit?

- a. How many strings of four hexadecimal digits do not have any repeated digits?

Let A be the set of all the strings of four hexadecimal digits which do not have any repeated digits. There're 16 hexadecimal digits.

Since there're four positions to fill, and sixteen ways to choose the first position. And since no digit is allowed to be used again, there're only fifteen ways to choose the second position, and fourteen ways to choose the third position, and then thirteen ways to choose the fourth position. Hence:

$$16 \times 15 \times 14 \times 13 = 43,680$$

Solution: a. There're 43,680 strings of four hexadecimal digits with no repeated digits.

- b. How many strings of four hexadecimal digits have at least one repeated digit?

Let S be the set of all strings of four hexadecimal digits with repetition allowed:

$$S = 16 \times 16 \times 16 \times 16 = 16^4 = 65,536$$

And let A be the set of hexadecimal digits where no repetition is allowed:

$$A = 16 \times 15 \times 14 \times 13 = 43,680$$

Then S - A is the set of four hexadecimal digits with at least one repeated symbol, and, by the difference rule,

$$\begin{aligned} N(S - A) &= N(S) - N(A) \\ &= 65,536 - 43,680 \\ &= 21,856. \end{aligned}$$

Solution: b. There're 21,856 strings of four hexadecimal digits with at least one repeated digit.

c. What is the probability that a randomly chosen string of four hexadecimal digits has at least one repeated digit?

Let $E \subseteq S$, where E is the set containing all strings of four hexadecimal strings with at least one repeated digit, and S is the set containing all strings of four hexadecimal digits, then:

$$P(E) = \frac{E}{S} = \frac{21,856}{65,536} = 0.3334960938 \approx 0.33$$

Solution: c. The probability of a randomly chosen string of four hexadecimal digits having at least one repeated digit is 33%.